

Notes: Deschênes, Greenstone, and Shapiro (AER, 2017)

Defensive Investments and the Demand for Air Quality: Evidence from the NOx Budget Program

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1 Big picture

- **Question.** What is the willingness to pay for cleaner air when people can also make costly defensive investments, such as buying medications, to protect themselves against pollution?
- **Main setting.** The paper studies the NO_x Budget Program (NBP), a cap-and-trade market that reduced summertime nitrogen oxides emissions from electricity generators and industrial boilers in the eastern and midwestern United States.
- **Core empirical idea.** The NBP provides three sources of quasi-experimental variation: NBP versus non-NBP states, before versus after market implementation, and summer versus winter. The market operated only during the ozone season, May 1 to September 30.
- **Main pollution result.** The NBP sharply reduced summertime NO_x emissions and lowered ambient ozone concentrations. The decline is visible in daily emissions and in event-study estimates.
- **Main defensive-investment result.** Medication expenditures fall after the NBP. The paper interprets this as evidence that households use medications as defensive investments against air pollution.
- **Main health result.** The NBP reduced mortality, especially among individuals aged 75 and older. The mortality effects are economically large, while hospital-cost effects are less clear.
- **Main welfare result.** Annual savings in medication purchases are about \$800 million, while monetized mortality benefits are about \$1.3 billion. Defensive investments therefore account for more than one-third of estimated willingness to pay for reductions in NO_x.
- **Interpretive contribution.** Standard pollution-health studies often measure the total derivative of pollution on health after people defend themselves. This paper also measures an important defensive margin, reducing downward bias in welfare estimates.
- **Takeaway.** Air-quality regulation generates benefits through both better health and reduced defensive expenditures. Ignoring defensive behavior understates the value of environmental improvements.

2 Incremental contribution of the paper

2.1 Contribution relative to pollution-health research

- Earlier quasi-experimental studies of pollution typically focus on mortality, hospitalizations, infant health, or other direct health outcomes.
- Those outcomes are affected after households have already chosen defensive actions, such as staying indoors, purchasing air filters, visiting doctors, or buying medications.
- This paper adds a direct empirical measurement of an important defensive margin: medication purchases.
- By measuring both health outcomes and defensive expenditures, the paper shows that the benefits of pollution reductions are larger than health-outcome estimates alone imply.

Interpretation: The incremental contribution is not only to estimate another pollution effect. It changes the welfare accounting by adding a previously hidden component of willingness to pay.

2.2 Contribution relative to defensive-investment theory

- The Grossman-style health-production framework predicts that individuals respond to harmful environments with costly compensatory actions.
- The empirical literature has long recognized this point but has rarely measured defenses directly.

- The paper provides a clean empirical example in which a policy-induced decline in pollution reduces spending on medicines.

Interpretation: This bridges a gap between theory and measurement. Defensive investments are a theoretically central but empirically undermeasured component of demand for environmental quality.

2.3 Contribution relative to emissions-market evaluation

- Market-based environmental policies are often evaluated using engineering abatement costs and modeled environmental benefits.
- The paper links an actual cap-and-trade market to emissions, ambient pollution, medication purchases, and mortality in the same empirical framework.
- It shows that the NBP's benefits plausibly exceed an upper-bound estimate of compliance costs.

Interpretation: The paper is an empirical cost-benefit evaluation of a large emissions market, using realized health and defense outcomes rather than only model-based projections.

2.4 Contribution relative to valuation of air quality

- The paper provides IV estimates of the effect of NO_x emissions and ozone on medication purchases and mortality.
- It separately monetizes medication-cost savings and mortality reductions.
- It shows that the medication component is large enough that ignoring it materially understates willingness to pay.

Interpretation: The paper reframes the demand for clean air as the sum of avoided illness, avoided mortality, and avoided defensive expenditures.

3 Conceptual framework: defensive investments and willingness to pay

3.1 Health production

Let s denote sick time, c denote ambient pollution, and a denote defensive investments. The health production function is

$$s = s(c, a). \quad (1)$$

Defensive investments are assumed to reduce sick time, so $\partial s / \partial a < 0$, while pollution increases sick time, so $\partial s / \partial c > 0$.

Interpretation: Health outcomes alone do not reveal the full effect of pollution because people can spend resources to reduce the observed health damage.

3.2 Total versus partial effects

The observed relationship between pollution and sick time reflects both direct biological effects and defensive behavior:

$$\frac{ds}{dc} = \frac{\partial s}{\partial c} + \frac{\partial s}{\partial a} \frac{\partial a^*}{\partial c}. \quad (2)$$

Because $\partial s / \partial a < 0$ and $\partial a^* / \partial c > 0$, observed health effects ds/dc can understate the direct harm $\partial s / \partial c$.

Interpretation: If pollution rises and households buy more medication, measured illness may rise only a little. The small health effect does not mean pollution is harmless; it may mean defensive behavior is costly and effective.

3.3 Willingness to pay

The marginal willingness to pay for cleaner air includes:

1. avoided productivity losses from sick time;
2. avoided defensive expenditures;
3. avoided disutility and mortality from sickness.

In a simplified expression,

$$WTP_c = \text{value of avoided sick time} + p_a \frac{\partial a^*}{\partial c} + \text{value of avoided health disutility and mortality.} \quad (3)$$

Interpretation: The second term is the paper’s focus. Medication costs are not the full universe of defensive investments, but they are a major observable component.

3.4 Medication costs as defenses

The paper treats medication purchases as a defensive or mitigating response to air pollution.

- Short-acting respiratory medications can respond to symptoms after exposure.
- Long-term control medications can prevent or reduce episodes before they occur.
- Cardiovascular medications can also be linked to pollution-related health effects.

Interpretation: From a welfare perspective, it does not matter whether the medication is averting exposure damage beforehand or mitigating symptoms afterward. Both are costly responses to pollution.

4 Institutional background and data

4.1 The NO_x Budget Program

- The NBP was a cap-and-trade program for NO_x emissions from power plants and industrial boilers.
- It operated during the ozone season, May 1 through September 30.
- The program began partially in 2003 and was fully in place in 2004 for the main participating states.
- The program was designed to reduce ozone, because NO_x is a key precursor of summertime ozone formation.

Interpretation: The seasonal nature of the policy is crucial. It creates within-year variation that helps separate the policy from other time-varying confounders.

4.2 Pollution data

- Emissions data come from power plants covered by the Acid Rain Program.
- Ambient pollution data come from EPA monitoring networks.
- Outcomes include NO_x emissions, SO₂, CO₂, ozone 8-hour values, high-ozone days, carbon monoxide, sulfur dioxide, nitrogen dioxide, and particulates.

- The key ambient outcome is ozone, measured both by concentration and by days above thresholds such as 65 ppb.

Interpretation: The NBP directly regulates NO_x , while ozone is formed by atmospheric chemistry. The direct first stage is therefore strongest for NO_x emissions, with ozone as an important downstream outcome.

4.3 Medication and hospitalization data

- Medication data come from MarketScan, a confidential insurance-claims database.
- The data include large employer-sponsored insurance records from 2001 to 2007.
- The paper observes medication purchase date, county, National Drug Code, and payments by insurer plus consumer.
- Medication categories are classified as respiratory/cardiovascular or other categories.
- Hospitalization data are also from MarketScan, although the paper finds less systematic evidence for hospitalization effects.

Interpretation: Claims data provide a rare measure of defensive behavior at the county-season-year level.

4.4 Mortality and weather data

- Mortality data come from vital statistics and cover the longer 1997-2007 period.
- Mortality outcomes are deaths per 100,000 population by cause and age group.
- Weather controls include temperature, precipitation, and dew point temperature, flexibly controlled using distributional bins.

Interpretation: Weather is central because ozone formation and mortality both depend strongly on temperature and humidity.

5 Empirical specifications

5.1 Triple-difference reduced form

The main reduced-form specification is a differences-in-differences-in-differences model:

$$Y_{cst} = \gamma_1 I(\text{NBP Operating})_{cst} + W'_{cst} \beta + \rho_{ct} + \eta_{st} + \nu_{cs} + \varepsilon_{cst}. \quad (4)$$

- c indexes counties, s indexes seasons, and t indexes years.
- Y_{cst} is pollution, medication costs, hospitalization costs, or mortality.
- $I(\text{NBP Operating})_{cst}$ equals one for NBP states in NBP operating years during summer; it equals 0.5 in 2003.
- ρ_{ct} are county-by-year fixed effects.
- η_{st} are season-by-year fixed effects.
- ν_{cs} are county-by-season fixed effects.

Interpretation: The identifying variation compares summer versus winter changes in NBP states relative to non-NBP states before and after the market begins.

5.2 Event-study model

To examine pre-trends and dynamics, the paper estimates:

$$Y_{cst} = \sum_{r=1997}^{2007} \alpha_r I(\text{NBP State} \times \text{Summer} \times \text{Year}_r)_{cst} + W'_{cst} \beta + \rho_{ct} + \eta_{st} + \nu_{cs} + \varepsilon_{cst}, \quad (5)$$

with the year 2002 normalized to zero.

Interpretation: The event studies test for differential pre-trends and show whether effects appear when the market begins.

5.3 Instrumental variables

The reduced-form NBP treatment is also used as an instrument for NO_x emissions or ozone:

$$\text{Pollution}_{cst} = \pi I(\text{NBP Operating})_{cst} + \text{Controls} + FE + u_{cst}, \quad (6)$$

$$\text{Outcome}_{cst} = \theta \widehat{\text{Pollution}}_{cst} + \text{Controls} + FE + e_{cst}. \quad (7)$$

Interpretation: The IV estimates convert the market variation into estimates of the effect of NO_x or ozone on medication purchases and mortality.

6 Identification and interpretation

6.1 Identifying assumptions

The triple-difference design requires that, absent the NBP, summer-minus-winter outcome changes in NBP states would have followed the same path as summer-minus-winter changes in non-NBP states after conditioning on fixed effects and weather.

- County-by-year fixed effects absorb annual local shocks.
- Season-by-year fixed effects absorb national seasonal shocks.
- County-by-season fixed effects absorb permanent county-specific seasonal differences.
- Weather controls absorb nonlinear temperature and humidity channels.

Interpretation: The design is powerful because it exploits policy timing, geography, and seasonality simultaneously.

6.2 Threats and responses

- **Other regulations:** other pollution rules could confound estimates if they changed differentially in NBP states, summers, and post years.
- **Spillovers:** ozone and NO_x can travel across state borders, so the paper excludes adjacent states from the comparison group.
- **Medication insurance coverage:** MarketScan is not nationally representative, so the analysis weights by MarketScan coverage and population where appropriate.
- **Seasonal mortality displacement:** mortality may be displaced from summer to later months rather than prevented. The paper discusses alternative year definitions and acknowledges limited power for life-expectancy effects.

Interpretation: The paper's strongest evidence is where the timing, seasonality, and pollution chemistry all align with the NBP's institutional design.

6.3 How to interpret medication effects

Medication reductions are interpreted as avoided defensive investments. They are not a complete measure of defensive behavior because people can also change time outdoors, use air filtration, move, or undertake other avoidance behavior.

Interpretation: The medication estimates are therefore a lower bound on defensive-investment benefits, not a full measure of all avoidance costs.

7 Tables and figures

7.1 Figure 1 and Table 1: NBP emissions pattern and summary statistics

Read:

- Figure 1 shows average total daily NO_x emissions in NBP states before and after the program.
- Before the market, emissions are relatively high during summer. After the NBP, summertime emissions fall sharply between May 1 and September 30.
- Table 1 reports mean summer pollution, weather, medication-cost, hospitalization, and mortality variables.
- The table shows that medication spending and mortality are large enough that even modest percentage reductions are economically meaningful.

Economic message: Figure 1 shows the policy first stage directly. Table 1 provides scale: the defensive-cost and mortality outcomes are economically large.



FIGURE 1. TOTAL DAILY NO_x EMISSIONS IN THE NBP-PARTICIPATING STATES

Notes: Figure 1 shows average total daily NO_x emissions in the NBP participating states in 2001–2002 and 2005–2007. These estimates are obtained from an OLS regression of NO_x emissions on six day-of-week indicators and a constant. The values in the graph equal the constant plus the regression residuals, so that the graph depicts fitted values for the reference category (Wednesday). Total daily NO_x emissions on y-axis are measured in thousands of tons. The sample includes emissions from all the Acid Rain Units. NBP participating states include: Alabama, Connecticut, Delaware, Illinois, Indiana, Kentucky, Maryland, Massachusetts, Michigan, Missouri, New Jersey, New York, North Carolina, Ohio, Pennsylvania, Rhode Island, South Carolina, Tennessee, Virginia, and West Virginia, and Washington, DC. The NBP operated only in northeastern states on May 1 of 2003, and expanded

Figure 1: daily NO_x emissions in NBP states

TABLE 1—MEAN SUMMER VALUES OF THE POLLUTION, WEATHER, AND HEALTH VARIABLES, 2001–2007

	Counties with data (1)	Mean (2)	Standard deviation (3)
<i>Pollution emissions (000's of tons/summer)</i>			
NO_x emissions	2,539	0.52	(1.99)
SO_2 emissions	2,539	1.50	(6.52)
CO_2 emissions	2,539	384	(1,299)
<i>Air quality (ambient pollution)</i>			
Ozone 8-hour value (ppb)	168	48.06	(9.28)
Ozone days ≥ 65 (ppb)	168	23.60	(22.64)
NO_2 (ppb)	110	11.45	(5.39)
CO (ppm)	125	0.44	(0.24)
$\text{PM}_{2.5}$ ($\mu\text{g}/\text{m}^3$)	298	13.33	(4.19)
PM_{10} ($\mu\text{g}/\text{m}^3$)	39	27.28	(6.26)
SO_2 (ppb)	150	3.26	(2.27)
<i>Weather</i>			
Temperature ($^{\circ}\text{F}$)	2,539	70.57	(5.79)
Precipitation (1/100")	2,539	11.47	(5.36)
Dew point temp. ($^{\circ}\text{F}$)	2,539	57.72	(7.95)
<i>Medication costs (\$ per person)</i>			
All	2,435	377.56	(338.66)
Copayment	2,435	79.10	(59.21)
Respiratory or cardio.	2,435	99.59	(108.74)
<i>Hospitalizations (\$ per person)</i>			
All	2,435	601.76	(2,503.76)
Respiratory or cardio.	2,435	150.29	(972.70)
<i>Mortality (deaths per 100,000 people)</i>			
All	2,539	399.07	(118.90)
Respiratory or cardio.	2,539	179.30	(68.77)

Notes: Medication and hospitalization costs are reported in 2015 dollars and deflated using the US CPI for urban consumers. Emissions, medications, and deaths are totals per summer. Ambient pollution and weather are mean summer values. Means are across counties (i.e., not weighted). All data are for the period 2001–2007.

Table 1: summary statistics

7.2 Figure 2 and Table 2: pollution impacts

Read:

- Figure 2, panel A, shows an event-study decline in NO_x emissions after 2003.
- Figure 2, panel B, shows fewer high-ozone summer days in the relevant ozone bins.
- Table 2 shows that the NBP reduces NO_x emissions by roughly 0.33 to 0.44 thousand tons per county-season-year depending on specification.
- Table 2 also shows reductions in ozone 8-hour values and ozone days above 65 ppb.

Economic message: The NBP clearly changes the regulated pollutant and also reduces ambient ozone, the health-relevant pollutant targeted by the program.

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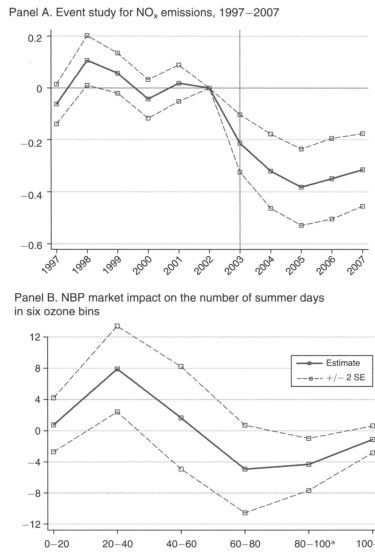


FIGURE 2. NBP MARKET IMPACT ON NO_x EMISSIONS AND AMBIENT OZONE POLLUTION

Notes: The estimates in Figure 2, panel A, are from an event study regression for NO_x emissions (measured in thousand of tons and observed at the county $\times$ year $\times$ season) where the estimates for year 2002 are restricted to have a value of 0. The regression includes detailed weather controls and a full set of county $\times$ year, season $\times$ year, and county $\times$ season fixed effects. The standard errors underlying the confidence intervals (dashed lines) are clustered at the state-season level. In panel B, the bins represent 6 categories of ozone 8-hour values, corresponding to the number of days per summer where the ozone 8-hour value is 0–20 ppb, 20–40 ppb, 40–60 ppb, etc. The ozone 8-hour value is measured as the maximum rolling 8-hour mean of hourly values within each day. The coefficients reported in Figure 2, panel B, are from a separate regression for each bin that includes detailed weather controls and a full set of county $\times$ year, season $\times$ year, and county $\times$ season fixed effects. The regression is weighted by the number of ozone readings in each county $\times$ season $\times$ year. The asterisk on the 80–100 category denotes the non-attainment air quality standard during the NBP years (85 ppb).

Figure 2: emissions and ozone event studies

TABLE 2—EFFECT OF THE NBP MARKET ON EMITTED AND AMBIENT POLLUTION

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Pollution emissions (000s of tons per summer)</i>					
1. NO _x	−0.36 (0.05)	−0.38 (0.06)	−0.36 (0.07)	−0.33 (0.07)	−0.44 (0.12)
2. SO ₂	−0.08 (0.04)	−0.14 (0.07)	−0.07 (0.05)	−0.07 (0.03)	−0.04 (0.12)
3. CO ₂	−2.66 (4.33)	−24.58 (15.89)	−3.56 (5.60)	−12.66 (6.48)	−83.05 (50.54)
<i>Panel B. Air quality (ambient pollution)</i>					
4. Ozone 8-hour value (ppb)	−2.92 (0.76)	−3.58 (0.86)	−2.87 (0.72)	−3.38 (0.56)	−3.37 (0.54)
5. Ozone days $\geq$ 65 (ppb)	−7.33 (2.44)	−8.28 (2.43)	−7.79 (2.77)	−9.55 (2.63)	−10.26 (2.46)
6. CO: carbon monoxide (ppm)	−0.05 (0.02)	−0.05 (0.04)	−0.05 (0.03)	−0.02 (0.02)	0.00 (0.02)
7. SO ₂ : sulfur dioxide (ppb)	0.16 (0.12)	0.25 (0.24)	0.13 (0.17)	0.16 (0.19)	0.32 (0.19)
8. NO ₂ : nitrogen dioxide (ppb)	−1.13 (0.21)	−0.18 (0.82)	−0.98 (0.38)	−0.78 (0.39)	−0.90 (0.51)
9. PM _{2.5} : particulates less than 2.5 micrometers (μ g/m ³)	—	—	—	−0.45 (0.32)	−1.03 (0.27)
10. PM ₁₀ : particulates less than 10 micrometers (μ g/m ³)	—	—	—	−1.20 (1.17)	−0.99 (1.48)
County-by-season fixed effects	X	X	X	X	X
Summer-by-year fixed effects	X	X	X	X	X
State-by-year fixed effects	X	X			
County-by-year fixed effects			X	X	X
Detailed weather controls		X	X	X	X
Data begins in 2001		X	X	X	X
Weighted by emission/pollution monitors (panel B only)	X	X	X	X	
Weighted by population					X

Notes: The entries in Table 2 are the coefficient estimates from the DDD estimator described in equation (4). Each coefficient is from a separate regression that includes a full set of county $\times$ year, season $\times$ year, and county $\times$ season fixed effects. Additional control variables are listed in the text. The reported standard errors are clustered at the state-season level. Emitted pollutant variables (panel A) are measured in thousand of tons and ambient pollutant variables (panel B) are mean values. Unless otherwise noted, the sample period begins in 1997. Ambient pollution regressions (panel B) are GLS weighted by the number of underlying pollution readings unless otherwise noted. For emissions, the number of observations is 55,838 in columns 1 to 3 and 35,546 for columns 4–5. For ambient pollution, the number of observations for each pollutant based on 1997–2007 sample is 3,146 (Ozone); 2,244 (CO); 2,684 (SO₂); 1,782 (NO₂). For ambient pollution based on 2001–2007 sample, the number of observations is 2,352 (Ozone); 1,750 (CO); 2,100 (SO₂); 1,540 (NO₂); 4,172 (PM_{2.5}); and 546 (PM₁₀). For emissions, share of population covered is 100 percent. For ambient pollution, share of population covered is 28–40 percent for ozone, CO, SO₂, and NO₂; 55 percent for PM_{2.5}; and 10 percent for PM₁₀.

comparison of the years 2000 and 2007. Reconciling this with our smaller estimate

Table 2: emitted and ambient pollution

7.3 Figure 3 and Table 3: defensive expenditures

Read:

- Table 3 shows that the NBP lowers medication costs per capita.
- The estimated decline is stronger for respiratory and cardiovascular medications, the class most plausibly related to air pollution.
- Figure 3, panel A, shows a post-NBP decline in respiratory and cardiovascular medication costs.
- Figure 3, panel B, previews the mortality response for respiratory and cardiovascular causes.

Economic message: This is the paper's key novel margin: cleaner air reduces costly defensive investments.

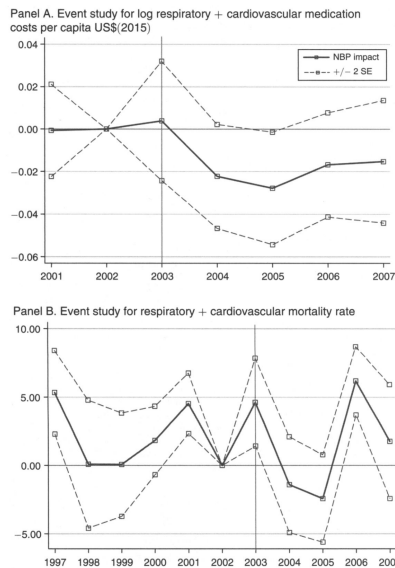


FIGURE 3. IMPACT OF THE NBP MARKET ON DEFENSIVE EXPENDITURES HEALTH OUTCOMES

Notes: The estimates in Figure 3, panel A, are from an event study regression for log medication costs per capita (respiratory and cardiovascular medications only), and the estimates in Figure 3, panel B, are from an event study regression for mortality rates per 100,000 population (respiratory and cardiovascular causes only). In both regressions, the estimates for year 2002 are restricted to have a value of 0. The regressions include detailed weather controls and a full set of county $\times$ year, season $\times$ year, and county $\times$ season fixed effects, and are weighted by county population (panel A uses MarketScan population and panel B uses total population). The standard errors underlying the confidence intervals (dashed lines) are clustered at the state-season level. Medication costs are in 2015 dollars, deflated using the BLS CPI for urban consumers.

Figure 3: medication and mortality event studies

TABLE 3—EFFECT OF THE NBP MARKET ON LOG MEDICATION COSTS PER CAPITA

	(1)	(2)	(3)	(4)
<i>Panel A. All medications</i>				
1. All medications	−0.008 (0.010)	−0.015 (0.008)	−0.016 (0.006)	−0.016 (0.007)
MarketScan as share of total population	0.007	0.007	0.007	0.002
Counties as share of total population	0.994	0.994	0.994	0.370
<i>Panel B. Specific types of medications</i>				
2. Respiratory or cardiovascular	−0.009 (0.013)	−0.019 (0.011)	−0.023 (0.008)	−0.019 (0.009)
3. Non-respiratory and non-cardiovascular	−0.009 (0.010)	−0.015 (0.008)	−0.015 (0.006)	−0.017 (0.007)
County-by-season fixed effects	X	X	X	X
Summer-by-year fixed effects	X	X	X	X
State-by-year fixed effects	X	X		
County-by-year fixed effects			X	X
Detailed weather controls		X	X	X
Only counties with ozone monitors				X
Weighted by population	X	X	X	X

Notes: Medication costs are reported in 2015 dollars and deflated using the US CPI for urban consumers. The entries in Table 3 are the coefficient estimates from the DDD estimator described in equation (4) when the dependent variable is the log of medication costs per person-season-year in a county. Each coefficient is from a separate regression that includes a full set of county $\times$ year, season $\times$ year, and county $\times$ season fixed effects. Additional control variables are listed in the bottom of Table 3. The reported standard errors are clustered at the state-season level. The regressions are GLS weighted by the MarketScan population in a given county-year-season. The reported standard errors are clustered at the state-season level. Total population refers to the 2,539 counties in the main sample. Number of observations is as follows: Row 1, columns 1–3: 30,926. Row 1, column 4: 2,338. Row 2, columns 1–3: 28,910. Row 2 column 4: 2,324. Row 3 columns 1–3: 30,730. Row 3 column 4: 2,338.

column 3 estimate in row 2 indicates that the NBP decreased expenditures on respiratory and cardiovascular medications by a statistically significant 2.3 percent. In the smaller sample of counties with ozone monitors in column 4, the point estimate

Table 3: medication costs

7.4 Table 4 and Table 5: mortality and IV estimates

Read:

- Table 4 shows mortality-rate reductions, especially among people age 75 and older.
- The all-age estimates are negative but not always precise, while the oldest group shows large reductions.
- Table 5 reports OLS and 2SLS estimates linking NO_x and ozone to medication purchases and mortality.
- The IV estimates imply that reductions in NO_x and ozone have substantial benefits for both medication spending and mortality.

Economic message: Mortality benefits and defensive-expenditure benefits both matter. The IV results translate the NBP treatment into pollutant-specific willingness-to-pay estimates.

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. All deaths</i>					
1. All deaths	−2.07 (0.94)	0.90 (3.58)	−1.70 (0.83)	−5.97 (1.77)	−3.03 (1.53)
<i>Panel B. Specific causes of death</i>					
2. Respiratory or cardiovascular	−0.71 (0.50)	0.45 (1.65)	−0.41 (0.67)	−2.42 (1.17)	−1.08 (0.95)
3. Non-respiratory and non-cardio.	−1.36 (0.57)	0.46 (2.15)	−1.29 (0.46)	−3.55 (0.79)	−1.95 (0.91)
4. External	0.32 (0.20)	0.22 (0.32)	0.22 (0.30)	−0.18 (0.62)	0.15 (0.38)
<i>Panel C. All causes of death, by age group</i>					
5. Age 0 (infants)	−1.66 (3.84)	−5.92 (6.06)	−5.48 (6.03)	−3.58 (10.61)	−11.64 (9.23)
Estimated change in 2005 deaths	−29	−102	−94	−62	−201
6. Ages 1–64	−0.05 (0.32)	−0.77 (0.82)	−0.26 (0.43)	−0.58 (1.13)	−0.56 (0.86)
Estimated change in 2005 deaths	−58	−897	−303	−676	−653
7. Ages 65–74	−3.16 (4.35)	−1.73 (10.22)	−3.40 (6.18)	−12.38 (9.94)	−6.61 (6.25)
Estimated change in 2005 deaths	−281	−154	−303	−1,103	−589
8. Ages 75+	−29.10 (8.14)	−2.37 (21.48)	−16.08 (9.82)	−82.56 (16.88)	−28.83 (15.91)
Estimated change in 2005 deaths	−2,496	−2,340	−1,275	−7,647	−1,509
County-by-season fixed effects	X	X	X	X	X
Summer-by-year fixed effects	X	X	X	X	X
State-by-year fixed effects	X	X	X	X	X
County-by-year fixed effects			X	X	X
Detailed weather controls		X	X	X	X
Counties with ozone monitors				X	
Data begins in 2001					X

Notes: The entries in Table 4 are the coefficient estimates from the DDD estimator described in equation (4) where the dependent variable is deaths per 100,000 population in each county-year-season. Each coefficient is from a separate regression that includes a full set of county $\times$ year, season $\times$ year, and county $\times$ season fixed effects. Additional control variables are listed in the bottom of Table 4. The reported standard errors are clustered at the state-season level. The regressions are GLS weighted by the relevant population in a given county-year-season. Unless noted otherwise, the data begin in 1997. Number of observations is 55,858 for columns 1–3; 3,146 for column 4; and 35,546 for column 5.

Table 4: mortality effects

7.5 Table 6: welfare impacts

Read:

- Table 6 compares NBP social costs with benefits from medication-cost reductions and mortality reductions.
- The upper-bound aggregate abatement cost is about \$1.1 billion per year.
- Medication-cost reductions are valued at about \$820 million per year.
- Mortality benefits are valued at about \$1.3 billion per year.
- Total benefits exceed the upper-bound cost estimate.

Economic message: Defensive investments are not a small add-on. They account for more than one-third of estimated benefits from the NBP.

TABLE 5—EFFECT OF NO_x EMISSIONS AND AMBIENT OZONE CONCENTRATIONS ON MEDICATION PURCHASES AND MORTALITY: OLS AND IV ESTIMATES

	log medication costs			All-cause mortality		
	All counties	Counties with NO _x emissions	Ozone monitored counties	All counties	Counties with NO _x emissions	Ozone monitored counties
	(1)	(2a)	(2b)	(3)	(4a)	(4b)
<i>Panel A. OLS</i>						
NO _x emissions	0.08 (1.13)	0.48 (1.06)	—	−0.02 (0.29)	−0.02 (0.29)	—
8-hour ozone	—	—	0.39 (0.86)	—	—	0.10 (0.15)
Days $\geq$ 65 ppb	—	—	0.20 (0.14)	—	—	−0.01 (0.03)
<i>Panel B. 2SLS</i>						
NO _x emissions	21.68 (13.59)	12.28 (5.77)	—	5.28 (3.65)	2.73 (1.84)	—
8-hour ozone	—	—	5.65 (2.05)	—	—	1.96 (1.01)
Days $\geq$ 65 ppb	—	—	1.54 (0.58)	—	—	0.69 (0.37)

Notes: The coefficient estimates in columns 1, 2a, and 2b are multiplied by 1000 for readability. Columns 1, 2a, and 2b are based on the 2001–2007 sample; columns 3, 4a, and 4b are based on the 1997–2007 sample. NO_x emissions are measured in thousand tons per county. All regressions include county $\times$ year, season $\times$ year, and county $\times$ season fixed effects, as well as the detailed weather controls. The reported standard errors are clustered at the state-season level. The regressions are GLS weighted by the relevant population in a given county-year-season (MarketScan or full population). In panel B, the endogenous variable is NO_x or ozone and the excluded instrument is Summer $\times$ Post $\times$ NBP interaction (see equation (6)). Number of observations is 30,926 for medication regressions including all counties, 7,616 for medication regressions including counties with NO_x emissions, 2,338 for medication regressions including only counties with ozone monitors, 55,858 for mortality regressions including all counties, 12,320 for mortality regressions including counties with NO_x emissions, and 3,146 for mortality regressions only including counties with ozone monitors. The sample of counties with NO_x emissions include those with nonzero emissions in the summer of 2002. The sample is smaller for medications than for mortality due to counties without no medication data or zero expenditures.

counties are somewhat larger. The estimates also indicate that a 1 ppb increase in the 8 hour ozone concentration or 1 additional day with a concentration exceeding 65 ppb lead to 1.96 and 0.69 additional summertime deaths per 100,000 people, respectively; correspondingly, a 10 percentage point increase in 8-hour ozone or in

Table 5: IV estimates

TABLE 6—THE WELFARE IMPACTS OF THE NBP AND THE SOCIAL BENEFITS OF NO_x AND OZONE REDUCTIONS

	Medication costs (\$ million) (1)	Medication copayments (\$ million) (2)	Mortality:		Total using (1) (\$ million) (5)	Total using (2) (\$ million) (6)
			Number of deaths (3)	Monetized value (\$ million) (4)		
<i>Panel A. An upper bound estimate of NBP's social costs</i>						
Upper bound per year	—	—	—	—	\$1,076	\$1,076
Upper bound, 2003–2007 total	—	—	—	—	\$4,843	\$4,843
<i>Panel B. Estimates of the NBP's benefits</i>						
Total per year	\$820	\$161	1,975	\$1,319	\$2,139	\$1,480
Total 2003–2007	\$3,690	\$725	8,887	\$5,935	\$9,625	\$6,660
<i>Panel C. The annual social benefits of NO_x reductions in NBP states (million tons)</i>						
Regressions based on . . .						
All counties	\$938	\$178	6,048	\$4,039	\$4,977	\$4,217
Counties with NO _x emissions	\$531	\$103	3,127	\$2,088	\$2,620	\$2,192
<i>Panel D. The social benefits of ozone reductions in NBP states (ppb)</i>						
1 ppb ozone decrease	\$290	\$41	2,660	\$1,777	\$2,066	\$1,818
1 Less day with ozone > 65 ppb	\$79	\$11	937	\$625	\$704	\$637

Notes: All dollar amounts are in 2015 constant dollars deflated using BLS CPI for urban consumers. The mortality impact estimates without dollar signs are number of deaths. The monetized mortality impact uses the VSL of \$2.27 million (2015 dollars) from Ashenfelter and Greenstone (2004) and the age adjustments from Murphy and Topel (2006, p. 888). The implied VSLs are as follows: \$2.26 million (infants); \$1.78 million (age 1–64); \$0.7 million (age 65–74); and \$0.3 million (age 75+). Total 2003–2007 decrease due to NBP assumes impact is for half of 2003 summer and for all of summers 2004–2007. NBP cost upper bound is based on the permit price of about \$2,523/ton US\$ (2015) and estimated total abatement quantity of 427,000 tons. The numbers in panel A comes from multiplying together the mean NBP allowance price per ton, the effect of the NBP on county-level NO_x emissions (Table 2, column 4), and the number of counties in the NBP states (1,185). The numbers in panels B–D come from multiplying together regression estimates of how the NBP, NO_x, or ozone affects medication costs or mortality by the total number of people in the NBP states in the year 2005 (136 million people). Specifically, panel B, column 1, uses the estimate from Table 3, column 3. Panel B, columns 2–3 use the estimates from Table 4, column 3, panel C. Panel C, column 1 uses the estimate from Table 5, column 1, panel B. Panel C, columns 2–3 use the estimate from Table 5, panel B, column 3. Panel D, column 1 uses the estimate from Table 5, panel B, column 2. Panel D, columns 2–3 use the estimate from Table 5, panel B, column 4. Panel D estimates are based on regressions using counties with ozone monitors. All estimates apply to the full population in NBP states. See the text for further details.

Table 6: welfare impacts and social benefits

8 Short summary

- The paper estimates the effect of the NO_x Budget Program on pollution, medication costs, and mortality.
- It uses a triple-difference design comparing NBP versus non-NBP states, post versus pre years, and summer versus winter.
- The NBP sharply reduces summertime NO_x emissions.
- The NBP also reduces ambient ozone and high-ozone days.
- Medication purchases fall, especially respiratory and cardiovascular medications.
- Mortality falls, with the largest effects among the elderly.
- IV estimates link reductions in NO_x and ozone to avoided medication purchases and deaths.
- Avoided medication costs are about \$800 million per year and mortality benefits are about \$1.3 billion per year.
- Defensive investments account for a large share of willingness to pay for clean air.
- The NBP's benefits plausibly exceed its costs.

9 Conclusion

9.1 Core conclusion

Deschênes, Greenstone, and Shapiro show that the demand for clean air cannot be measured only by mortality or hospitalization effects. Households also make costly defensive investments. When an emissions market lowers pollution, these defensive costs fall.

9.2 Economic intuition

Pollution harms health, but households can partly protect themselves. Observed health outcomes therefore reflect both the underlying harm of pollution and the costly actions taken to avoid that harm. Measuring medication purchases reveals part of this hidden cost.

9.3 Policy takeaway

The NOx Budget Program generated sizable benefits through both avoided mortality and avoided medication purchases. Environmental cost-benefit analyses that ignore defensive expenditures can materially understate the value of pollution control.